

INVESTMENTS

Ninth Canadian Edition · Bodie, Kane, Marcus

CHAPTER 2

Financial Markets, Asset Classes, and Financial Instruments

Verified against source text — learning objectives, formulas, and worked examples checked directly against the chapter.

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Study Guide · Prepared for Personal Use

LO1 · The Money Market

The money market is a subsector of the fixed-income market consisting of **short-term, highly marketable debt securities**. Most trade in large denominations, out of reach for individual investors directly — but money market mutual funds pool small investors' funds to buy a diversified basket of these instruments.

Instrument	Key Features
Treasury Bills (T-bills)	Issued by the Government of Canada. Zero default risk; the safest and most marketable Canadian money market instrument. Sold at a discount to face value, in denominations from \$1,000 up to \$1 million. Quoted as a bond equivalent yield (365-day year).
Certificates of Deposit (CDs) / GICs	A time deposit with a chartered bank; funds cannot be withdrawn on demand. In Canada, the small-investor version is called a Guaranteed Investment Certificate (GIC). Most CDs/GICs are non-transferable, but deposits over \$100,000 may be issued as negotiable Bearer Deposit Notes (BDNs) that can be resold before maturity. CDs/GICs are insured up to \$100,000 by the Canada Deposit Insurance Corporation (CDIC).
Commercial Paper	Unsecured short-term promissory notes from large, creditworthy corporations, often backed by a bank line of credit. Maturities up to one year (longer would require securities-act registration), typically under two months, in minimum denominations of \$50,000. Rated for credit quality; small investors access it only via money market funds.
Asset-Backed Commercial Paper (ABCP)	Short-term paper issued by financial firms (not non-financial corporations) and collateralized by other assets — in the mid-2000s, often subprime mortgages. The Canadian ABCP market froze in 2007 when investors fled and backup liquidity providers refused to honour their agreements, leaving roughly \$32 billion in paper unable to be rolled over. It was restructured into longer-term bonds by 2009 in a rescue effort involving future Bank of Canada governor Mark Carney.
Bankers' Acceptances (BAs)	Starts as an order to a bank by its customer to pay a sum at a future date (like a postdated cheque); the bank "accepts" it for a fee, assuming payment responsibility. Second only to T-bills in default safety. Widely used in foreign trade, and tradeable in the secondary market.
Repurchase Agreements (Repos)	A dealer sells government securities and agrees to buy them back the next day (or after a few days) at a slightly higher price; the price difference is the repo rate. Effectively a short-term, collateralized loan. A reverse repo is the mirror-image transaction.
Eurodollars	U.S. dollar-denominated deposits held at foreign banks or foreign bank branches, not subject to U.S. Federal Reserve regulation. Mostly large, short-term (under 6 months) time deposits. The Eurodollar CD variant can be resold before maturity.
Federal Funds /	Overnight loans of excess reserves between banks. In the U.S. this is the

Instrument	Key Features
Overnight Rate	federal funds rate; in Canada, the equivalent policy benchmark is the Bank of Canada's overnight rate.
LIBOR	The London Interbank Offered Rate — the rate at which large international banks lend to one another. Historically a key reference for floating-rate loans and derivatives, but the rate-fixing scandal of 2012 (banks colluding to manipulate their submissions) badly damaged its credibility, leading to its eventual discontinuation and replacement by alternative reference rates.

Key insight: money market instruments are "cash equivalents" — liquid and low-risk enough to convert to cash quickly with little loss in value. The 2008 crisis showed even this corner of the market is not risk-free: the Reserve Primary Fund "broke the buck" after holding Lehman commercial paper, and Canada's own 2007 ABCP freeze showed the same vulnerability domestically.

LO2 · The Bond Market

The bond market trades longer-term borrowing instruments than the money market: Government of Canada bonds, provincial and municipal bonds, corporate bonds, and mortgage securities. Collectively these make up the **fixed-income capital market**, since most promise either a fixed income stream or one determined by a specific formula.

Government of Canada Bonds & Notes

Direct federal obligations with essentially zero default risk in Canadian dollars; pay semi-annual coupons. Serve as the benchmark risk-free yield curve for the Canadian market. Some are callable, allowing the issuer to redeem at par (or a stated premium) before maturity.

Corporate Bonds

Let private firms borrow directly from the public, structured like government bonds (semi-annual coupons, face value at maturity) but carrying meaningful **default risk**. Three risk tiers by collateral: **secured bonds** (specific collateral backs the bond), **debentures** (unsecured), and **subordinated debentures** (lower-priority claim in bankruptcy). Corporate bonds often carry embedded options: **callable** (issuer may repurchase at a stated call price), **retractable** (holder may redeem early), **extendible** (holder may extend past stated maturity), and **convertible** (holder may convert into a set number of common shares).

Municipal Bonds — Important Canada/U.S. Distinction

In the **United States**, municipal bond interest is exempt from federal (and often state/local) income tax, making the equivalent taxable yield formula (Formulas page) central to comparing munis with taxable bonds.

In **Canada**, **municipal bonds are fully taxable** — there is no equivalent federal tax exemption. Canadian munis represent under 5% of the domestic fixed-income market (~\$35 billion outstanding as of 2018) and typically offer higher yields than comparable federal or provincial bonds with low default risk, but their appeal is limited by full taxability compared to their U.S. counterparts. (By contrast, over US\$3.8 trillion of U.S. municipal bonds were outstanding as of 2018 — more than the entire Canadian fixed-income market.)

General obligation bonds are backed by the issuer's full taxing power; **revenue bonds** are backed only by the cash flows of a specific project (airports, hospitals, turnpikes) and are riskier. An **industrial development bond** is a revenue bond issued to let a private firm access tax-exempt-style borrowing for a commercial project such as a factory.

Mortgage-Backed Securities (MBS)

An MBS is an ownership claim (or obligation) backed by a pool of mortgages — a **pass-through**: the originator continues servicing the loans but passes principal and interest payments to the security holder. First introduced in Canada by the Canada Mortgage and Housing Corporation (CMHC) in 1987; CMHC pass-throughs carry a federal guarantee under the National Housing Act, making their cash flows effectively risk-free even if individual borrowers default. In the U.S., agency MBS (Fannie Mae, Freddie Mac) carry similar guarantees, while private-label (subprime) pass-throughs do not — investors bear the default risk directly.

Real Return Bonds (Canada) / TIPS (U.S.)

Principal is indexed to the Consumer Price Index, so both principal and coupon payments rise with inflation. Provides a guaranteed real (inflation-adjusted) rate of return. The Canadian government issues Real Return Bonds (RRBs); the U.S. equivalent is Treasury Inflation-Protected Securities (TIPS).

LO3 · Equity Securities

Common Stock

Each share entitles its owner to one vote on corporate governance matters and a share in the firm's financial benefits. Shareholders elect a board of directors, which selects and oversees management — but management usually collects the vast majority of shareholder proxies and so retains considerable day-to-day discretion.

Two defining features: **residual claim** (paid last, after tax authorities, employees, suppliers, bondholders, and other creditors) and **limited liability** (shareholders can lose at most their original investment — unlike owners of unincorporated businesses, whose personal assets can be seized by creditors).

Some Canadian firms issue **dual-class shares** — e.g., Akita Drilling's Class A (multiple votes) versus Class B (single vote) shares are otherwise economically identical. The loss of voting power is generally reflected in a lower market value for restricted/lower-vote shares.

Stock Market Listings — Reading the TSX

Daily TSX summaries report ticker, 52-week high/low, volume, closing price, and change from prior close, plus dividend yield and P/E ratio where applicable. The TSX Composite Index tracks about 240 of the largest TSX-listed securities by market value.

Preferred Stock

A hybrid of debt and equity: pays a **fixed dividend** like a bond coupon (economically resembling a perpetuity) and carries no vote, but failure to pay does *not* trigger bankruptcy — unpaid dividends simply accumulate (**cumulative preferred**) and must be paid in full before any common dividend. Unlike bond interest, preferred dividends are **not tax-deductible** for the issuing firm. This is partly offset because corporations may exclude 70% of dividends received from other domestic corporations from taxable income — making preferred stock a relatively attractive fixed-income-like holding for *corporate* investors specifically (individual investors cannot use this exclusion, so preferred yields often look unattractive to them relative to bonds). Variants include callable/redeemable, convertible, and adjustable-rate preferred stock.

Depository Receipts

American Depository Receipts (ADRs) are certificates traded in U.S. markets representing ownership in shares of a foreign company (one ADR may equal a fraction, one, or several foreign shares). They let U.S. investors trade foreign equities without dealing directly in a foreign market or currency, and are the most common vehicle for U.S. investment in non-U.S. companies.

LO4 · Stock and Bond Market Indexes

Market indexes aggregate many security prices into one number to track overall performance. Three weighting methodologies exist:

Index Type	Methodology & Key Points
Price-Weighted (Dow Jones Industrial Avg.)	Each stock contributes in proportion to its share price, so a high-priced stock dominates regardless of the firm's total market size. Computed as the sum of prices divided by a divisor; the divisor is adjusted down after stock splits or component substitutions so the index stays continuous through those events.
Value-Weighted (S&P/TSX Composite, S&P 500)	Each stock's weight equals its market capitalisation (price × shares outstanding) as a share of total index market cap. Tracks the return on a portfolio held in proportion to market value, and is automatically unaffected by stock splits (both price and share count change, but market cap does not).
Equally Weighted	Every stock gets the same weight regardless of price or size — equivalent to investing equal dollar amounts in each and rebalancing periodically. Gives more relative influence to small-cap names than value-weighted methods.

The S&P/TSX Composite — Canada's Benchmark

Canada's best-known stock indicator: a value-weighted index currently covering roughly 240 of the largest TSX-listed securities by market value, across all industries, but excluding controlling blocks over 20% of outstanding shares.

Major Canadian & U.S. Indexes

Index	Market	Weighting	Description
S&P/TSX Composite	Canada	Value-weighted	Primary Canadian benchmark (~240 largest TSX stocks)
S&P 500	U.S.	Value-weighted	500 largest U.S. companies; primary U.S. professional benchmark
Dow Jones Industrial Avg.	U.S.	Price-weighted	Oldest U.S. index (30 blue-chip stocks); widely cited but narrow coverage
NASDAQ Composite	U.S.	Value-weighted	All NASDAQ-listed stocks; heavily technology-weighted
Wilshire 5000	U.S.	Value-weighted	Broadest U.S. index; despite the name, covers ~6,000 stocks

Bond Market Indexes

Harder to compute reliably than equity indexes: many bonds trade infrequently, so up-to-date market prices are often unavailable and must be estimated from bond-valuation models ("matrix prices"), which can diverge from true market values.

LO5 · Derivative Markets

Options and futures provide payoffs that depend on (are **contingent on**) the value of another asset — a commodity, bond, stock, or index. Used to hedge risk or to speculate.

Options

Type	Right Granted	Pays Off When
Call Option	Right to BUY the underlying at the exercise (strike) price on or before expiration	Underlying price RISES above the strike price
Put Option	Right to SELL the underlying at the exercise (strike) price on or before expiration	Underlying price FALLS below the strike price

The buyer pays a premium for the right; maximum buyer loss is the premium paid. The writer's potential loss can be far larger — theoretically unlimited for an uncovered call writer.

Futures Contracts

A futures contract is an **obligation** (not a right) to buy or sell at a stipulated futures price on a maturity date. The **long** position (agrees to buy) gains as the asset price rises; the **short** position (agrees to deliver) gains as the asset price falls. Futures are marked to market daily — gains/losses settle into margin accounts each day rather than only at expiry — which greatly reduces counterparty credit risk relative to OTC forward contracts, which are not marked to market.

Options vs. Futures: an option buyer has a right (capped loss = premium); a futures participant has an obligation (losses can exceed any initial payment). Both provide leverage but with fundamentally different risk profiles.

The Capital Markets Landscape — Connecting All Instruments

Chapter 2 supplies the vocabulary every later chapter builds on. Instruments positioned by maturity and default risk:

	Near-Zero Default Risk	Low–Moderate Credit Risk	Higher Credit Risk
Short-term (≤ 1 yr)	T-bills, BAs (money market)	Commercial paper, ABCP, repos	High-yield commercial paper (rare)
Medium–long-term (1–30 yr)	Canada bonds, RRBs, provincials	Investment-grade corporates, MBS	High-yield ("junk") bonds, ABS
No maturity (perpetual)	—	Preferred stock	Common stock

Key Relationships to Remember

- **Price and yield move inversely** for bonds — rising market yields push existing bond prices down.
- **Yield premiums over T-bills** compensate for credit, liquidity, and maturity risk — each incremental risk has a price.
- **Canadian munis are fully taxable** (unlike U.S. munis), which limits their appeal versus federal/provincial bonds despite higher stated yields.
- **Price-weighted indexes** are dominated by high-priced stocks and are altered by splits; value-weighted indexes are immune to splits and reflect true market-portfolio performance.
- **Derivatives provide leverage** without large upfront capital — useful for hedging, but able to amplify losses quickly.
- **Securitisation** (MBS/ABS/CDOs) converts illiquid loans into tradeable securities, expanding credit access — but can misalign originator incentives if the originator no longer bears the credit risk.

Chapter 2 — Big-Picture Takeaways

- Money markets (short-term, low-risk) and capital/bond markets (longer-term, more price-sensitive) together make up the fixed-income universe; equity and derivatives sit alongside them.
- Government T-bills anchor the risk-free end of the yield spectrum; every other instrument is priced at a spread above them reflecting its incremental risks.

- Canada-specific instruments and rules (CDIC insurance limits, CMHC pass-throughs, fully taxable munis, Real Return Bonds, the 2007 ABCP crisis) differ meaningfully from their U.S. counterparts.
- Index construction methodology matters: price-weighted and value-weighted indexes track different implicit portfolios and respond differently to stock splits.
- Derivatives are risk-transfer tools whose power (and danger) comes from leverage.

These are regularly tested in problem sets and exams. Know both what each computes and why it is structured that way.

F1 | Bond Equivalent Yield (Canada — 365-day year)

$$r_{\text{BEY}} = [(\text{FaceValue} - \text{Price}) / \text{Price}] \times (365 / n)$$

Price = current T-bill market price; n = days to maturity. Annualises via simple interest on a 365-day year (the Canadian convention). Understates the true effective annual return because it does not compound.

F2 | T-Bill Price from Bond Equivalent Yield

$$\text{Price} = \text{FaceValue} / [1 + r_{\text{BEY}} \times (n / 365)]$$

Rearrangement of F1. For a \$1,000 face bill with 91-day BEY = 0.99%: Price = 1000 / [1 + 0.0099 × (91/365)] = \$997.53.

F3 | Effective Annual Yield (EAY) on a T-Bill

$$\text{EAY} = (1 + \text{HPR})^{(365/n)} - 1, \text{ where } \text{HPR} = (\text{FaceValue} - \text{Price}) / \text{Price}$$

HPR = holding-period return for the actual days held; EAY compounds it to a full year. Always ≥ BEY since BEY uses simple interest.

F4 | U.S. Bank Discount Yield (360-day year)

$$d = [(\text{FaceValue} - \text{Price}) / \text{FaceValue}] \times (360 / n)$$

$$\text{Conversion: } r_{\text{BEY}} = (365 \times d) / (360 - d \times n)$$

Note the denominator is face value (not price) and the year is 360 days — both make the U.S. quote lower than the equivalent Canadian BEY for the same bill.

F5 | Equivalent Taxable Yield (U.S. Municipal Bonds)

$$r_{\text{taxable}} = r_{\text{muni}} / (1 - t)$$

$$\text{Cutoff tax rate: } t^* = 1 - (r_{\text{muni}} / r_{\text{taxable}})$$

Applies to U.S. municipal bonds, which are tax-exempt. Canadian municipal bonds are fully taxable, so this formula is not used to evaluate them — compare Canadian munis to other taxable bonds directly on yield.

F6 | Price-Weighted Index

$$\text{Index Value} = (\sum \text{Stock Prices}) / \text{Divisor}$$

The divisor starts at the number of stocks and is adjusted down after splits/component changes to keep the index continuous.

F7 | Value-Weighted (Market-Cap) Index

$$\text{Index}_t = \text{Index}_{\text{base}} \times (\sum \text{Market Cap}_t) / (\sum \text{Market Cap}_{\text{base}})$$

$$\text{Market Cap} = \text{Price} \times \text{Shares Outstanding}$$

Tracks a portfolio held in proportion to market cap; automatically unaffected by stock splits.

EXAMPLE A | T-Bill Pricing — BEY vs. EAY

Given: A \$1,000 face value Canadian T-bill, priced at \$960, 182 days to maturity.

Step 1 — BEY:

$$r_{\text{BEY}} = [(1000 - 960)/960] \times (365/182) = [40/960] \times 2.0055 = 0.08356 = 8.356\%$$

Step 2 — HPR:

$$\text{HPR} = (1000 - 960)/960 = 40/960 = 0.04167 = 4.167\%$$

Step 3 — EAY:

$$\text{EAY} = (1.04167)^{(365/182)} - 1 = 0.0851 = 8.51\%$$

Why EAY > BEY: BEY uses simple interest ($\times 365/n$); EAY compounds the holding-period return. Compounding on compounding makes EAY larger; the gap narrows for shorter bills and widens for longer ones.

EXAMPLE B | Price-Weighted vs. Value-Weighted Index

Setup: ABC: \$25 → \$30, 20M shares. XYZ: \$100 → \$90, 1M shares.

Stock	P ₀	P ₁	Shares (M)	Return
ABC	\$25	\$30	20M	+20%
XYZ	\$100	\$90	1M	-10%

Price-weighted:

$$\text{Initial} = (25 + 100)/2 = 62.5; \quad \text{Final} = (30 + 90)/2 = 60.0 \rightarrow \text{Return} = -4.0\%$$

XYZ dominates because it has the higher price, so its \$10 decline outweighs ABC's \$5 gain, despite ABC's larger percentage gain.

Value-weighted (base = 100):

$$\text{Initial cap: } 500\text{M} + 100\text{M} = 600\text{M}; \quad \text{Final cap: } 600\text{M} + 90\text{M} = 690\text{M} \rightarrow \text{Index} = 100 \times (690/600) = 115 \rightarrow \text{Return} = +15.0\%$$

ABC dominates here because it has the larger market cap (\$500M vs. \$100M) — its 20% gain outweighs XYZ's 10% loss when weighted by size.

PRACTICE PROBLEM | NorthStar Corp. Portfolio Manager

Scenario

You are a junior portfolio manager at NorthStar Investment Management in Calgary, analyzing a short-term portfolio and an index-tracking problem.

Money Market Holdings:

Instrument	Price Paid	Days to Mat.	Notes
Canadian T-bill (A)	\$989.50	91 days	Standard Govt. of Canada bill
U.S. T-bill (B)	\$982.00	180 days	Quoted as bank discount yield (360-day)
Commercial Paper (C)	\$994.00	60 days	Rated A-1; backed by bank credit line
Ontario Muni Bond (D)	4.20% coupon; priced at par	5 years	Fully taxable in Canada (unlike U.S. munis)

Equity Index Data:

Stock	Start	End	Shares Out.	Notes
RockFord Energy	\$80	\$92	5 million	No corporate actions
MapleGrow Ltd.	\$40	\$36	30 million	2-for-1 split at start of period

Part A — T-Bill Yields (Canadian Bill)

For Canadian T-bill (A): price \$989.50, face \$1,000, 91 days.

- Calculate the bond equivalent yield (BEY). Show your work.
- Calculate the effective annual yield (EAY). Show your work.
- In one sentence, explain why $EAY > BEY$.

Part B — U.S. Bank Discount Yield Conversion

For U.S. T-bill (B): price \$982.00, face \$1,000, 180 days.

- Calculate the U.S. bank discount yield (d), 360-day convention.
- Convert d to the equivalent Canadian BEY.

(iii) Why is the U.S. bank discount yield lower than the Canadian BEY for the same bill?

Part C — Municipal Bond Tax Treatment (Canada vs. U.S.)

The Ontario municipal bond (D) offers a 4.20% yield and is fully taxable in Canada.

- (i) Your client has a 45% marginal tax rate. What is their after-tax return on bond D?
- (ii) A comparable provincial bond offers 5.10% (also taxable). Which should the client prefer, and why?
- (iii) Would the equivalent-taxable-yield formula (F5) help compare these two bonds? Explain why or why not.

Part D — Price-Weighted Index

RockFord Energy and MapleGrow Ltd. form a price-weighted index.

- (i) MapleGrow splits 2-for-1 at the START of the period. Adjust the divisor so the index is unchanged at the split date. Show the calculation.
- (ii) Using the adjusted divisor and end-of-period prices, calculate the index return.
- (iii) What was the actual return on a portfolio holding one share of each stock (pre-split)?

Part E — Value-Weighted Index

Using the same two stocks, compute a market-value-weighted index (base = 100).

- (i) Compute start- and end-of-period total market cap for each stock.
- (ii) Calculate the index level at period end and the index return.
- (iii) Why does the value-weighted index give a different return than the price-weighted index?

Part F — Instrument Classification & Risk

Classify each instrument in the scenario table (A, B, C, D):

- (i) Money market vs. capital market.
- (ii) Primary risk(s) borne by the investor (credit, interest-rate, liquidity, prepayment, or none).
- (iii) Rank A, B, and C from lowest to highest yield, explaining the ranking via risk premiums.

Hints & Guidance

Parts A–B test F1–F4. Part C tests the Canada/U.S. municipal distinction directly — note that F5 does not apply to a fully-taxable Canadian muni. Parts D–E test F6–F7 (remember: the split happens at the START for Part D, so adjust the divisor before prices move; the split is irrelevant for Part E). Part F: T-bills carry zero credit risk, commercial paper carries slight credit risk, and the muni carries interest-rate risk from its 5-year maturity.